

WTMPI — Project Record

WildThread Modular Pack Interface

Developed by Artisans Wildthread

This document is a shareable project record compiled from the conversation and project decisions available in this chat. It is intended as a convenient reference and starting point for publishing the project. It is not a verbatim export of the ChatGPT application conversation.

1. Project Concept

WTMPI (WildThread Modular Pack Interface) is an open-source hardware platform for integrating modular electronics with backpacks and other soft goods. The architecture separates the reusable electronic interface from any single backpack implementation.

The initial reference implementation explores:

- ESP32-based external display module
- Removable clip-on electronics enclosure
- Backpack-integrated cable routing
- Weather-resistant external connection
- Internal USB/data interface
- Connection to a tablet or other primary computing device

2. Modular Architecture

The WTMPI concept is intended to support interchangeable modules while maintaining a consistent physical and electrical integration approach.

- **WTMPI Mount** — standardized physical attachment
- **WTMPI Display** — ESP32 display module
- **WTMPI Power** — power distribution module
- **WTMPI Sensor** — environmental and other sensors
- **WTMPI Light** — lighting/safety module
- **WTMPI Hub** — internal USB/data interface
- **WTMPI Backpack Integration** — cable routing and soft-goods integration

3. Reference Electronics

The first hardware concept uses a Waveshare ESP32-C6 1.47-inch display development board with a 172 × 320 display. The board is treated as a reference module rather than a definition of the entire WTMPI platform.

Potential functions discussed include battery/charging status, connectivity status, time, navigation/heading information, temperature, media controls, notifications, storage/device status, USB/power status, and other custom backpack information.

4. Mechanical Integration

The concept uses a 3D-printed external shell with a clip or mount for backpack webbing. A protected cable path leads through the backpack to an internal connector and computing/power interface. The preferred design direction emphasizes removable, serviceable connections rather than permanently loose wiring.

Conceptual enclosure illustrations explored weather-resistant construction, TPU-style protective elements, a rigid frame, sealed/bulkhead-style connectors, locking mounting features, rubber pads, and small mechanical fasteners. Any dimensions shown in early concept images were illustrative and should not be treated as verified production dimensions.

5. Reference Backpack

The current development platform is an experimental 10-liter backpack implementation. The project explored a colorway inspired by the REI Link 10 Hydro and rugged construction ideas associated with the Burton Incline family. These references are design inspiration and should not be represented as endorsement or affiliation.

The backpack implementation remains an example of WTMPI integration; WTMPI itself is intended to remain a broader modular interface architecture.

6. Open-Source GitHub Repository

A public GitHub repository was created under the Artisans Wildthread account:

Repository: WTMPI

Description: WildThread Modular Pack Interface — an open-source hardware platform for integrating modular electronics with backpacks and other soft goods.

The repository is public, uses the main branch, and contains an initial README establishing the project identity, architecture, reference implementation, project status, and open-source intent.

7. Current README Identity

WTMPI

WildThread Modular Pack Interface

An open-source hardware platform for integrating modular electronics with backpacks and other soft goods.

The README describes WTMPI as a modular interface architecture for physically and electrically integrating electronic devices with backpacks, bags, and outdoor equipment, with removable modules and a consistent approach to mounting, connection, cable routing, and integration.

8. Planned Repository Structure

- **docs/** — Technical documentation, architecture, interfaces, specifications, assembly, and revisions.
- **hardware/** — Enclosures, mounting clips, connector hardware, mechanical drawings, and exports.
- **electronics/** — Wiring diagrams, electrical architecture, connector pinouts, schematics, and component information.
- **firmware/** — ESP32 and other embedded software.
- **cad/** — CAD source files and related mechanical design assets.
- **backpack-integration/** — Cable routing, internal interfaces, attachment, and soft-goods integration.
- **patterns/** — Printable sewing patterns and related pattern documentation.
- **images/** — Project imagery and visual documentation.
- **bom/** — Bill of materials and component lists.

9. Immediate Next Step

The next repository task is to create **docs/README.md**. The planned contents introduce the documentation directory and identify system architecture, mechanical interface, electrical interface, module specifications,

backpack integration, assembly procedures, and design revisions as future documentation areas.

10. Licensing Status

No license has been selected yet. This was deliberate because WTMPI is expected to contain a mixture of hardware design files, CAD, documentation, sewing patterns, firmware, and other project assets. Licensing should be chosen deliberately for the different types of work rather than applying a generic software license to everything.

11. Canonical Project Source

The intended publishing model is: Instagram and similar social platforms serve as the showcase and discovery layer, while the public GitHub WTMPI repository becomes the canonical project source containing the authoritative/current design files and documentation.

This structure allows WTMPI to grow beyond the first ESP32 display and first 10-liter backpack implementation without requiring the overall project identity to change.

12. Project Identity

WTMPI

WildThread Modular Pack Interface

Project type: Open-source hardware / modular backpack electronics platform

Developer identity: Artisans Wildthread